

REVIEW

What the body does when Ozempic stops

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The last injection is the start of another phase, as appetite and blood-sugar effects fade. Studies find substantial weight regain, but their averages leave room for people who keep part of the loss—and for unanswered questions about how.

At the end of a large semaglutide trial, the injections stopped—and the researchers kept watching.¹ For more than a year, adults with overweight or obesity had received weekly semaglutide or a dummy injection, alongside regular counselling about food and physical activity.¹ A smaller group agreed to return after treatment finished, so investigators could record their weight, blood pressure and blood tests over the following year.¹ The scale began to tell a different story: much of the weight lost on the medicine came back.¹

That follow-up, the STEP 1 extension, found that participants previously taking semaglutide regained about two-thirds of their earlier weight loss within a year.¹ Many improvements in blood sugar, blood pressure and other measures of health moved back toward their starting levels too.¹ Yet the average concealed a second finding: nearly half of those with a final weight measurement remained at least about five per cent lighter than before treatment.¹

Benefits often fade after stopping, and substantial regain is common; complete regain is not inevitable.^{1,2} There is also a crucial distinction behind that answer: the trial used a weekly dose of 2.4 mg in people without diabetes.¹ Ozempic and the obesity medicine Wegovy contain the same active ingredient, semaglutide, but were developed for different purposes—type 2 diabetes and obesity management, respectively.³ STEP 1 studied people

without diabetes at the obesity dose, and STEP 4 did too; those participants differ from Ozempic users treated for diabetes.^{1,2}

01 MUCH COMES BACK, SOME STAYS OFF

The STEP 1 investigators followed 327 participants from selected trial sites who had completed semaglutide or placebo treatment.¹ Everyone in this follow-up stopped their assigned treatment.¹ Both the injections and the trial's lifestyle counselling ended together.¹ That matters: the rise in weight followed the loss of a treatment package, so the study leaves the drug's share of the effect uncertain.¹

The former semaglutide group lost an average 17.3% during treatment and remained 5.6% below its starting weight a year after stopping (Figure 1).¹ For a clearly hypothetical 100 kg starting weight, those two percentages calculate to about 17 kg lost during treatment and about 6 kg still lost a year afterward. This arithmetic illustrates the group averages.

A smaller lasting loss can still matter for health.² The STEP 4 paper cites guidance recommending sustained losses of about 5% to 15% to improve many conditions linked to overweight and obesity.² In the STEP 1 follow-up, nearly half of the former semaglutide participants with a final measurement still met the lower end of that

range.¹ Some had lost much more during treatment than others, and the average regain did not send everyone to the same endpoint.¹

Looking further ahead is harder.⁴ A review combining weight-management studies estimated that regain after newer medicines, semaglutide and tirzepatide, averaged a little under a kilogram a month.⁴ Its model projected a re-

turn to starting weight around a year and a half after stopping.⁴ Most of the studies followed patients for much shorter periods, so that date extends a model beyond the observed results.⁴ The measured year after STEP 1 is firmer ground than a forecast of what must happen next.^{1,4}

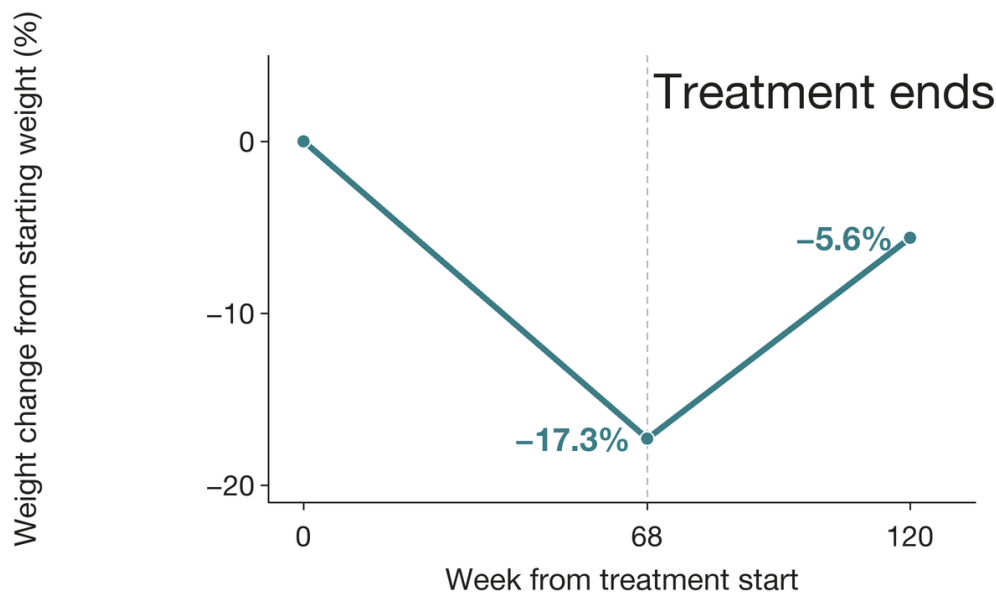


Figure 1. Average weight rose after treatment ended. The horizontal axis is weeks from treatment start; the vertical axis is percentage change from starting weight. The dots show selected group means without intervals, and connecting segments are visual guides rather than measured intermediate paths. STEP 1 ended both injections and counselling;¹ STEP 4 retained counselling during its randomized withdrawal comparison.²

02 HOW THE MEDICINE CHANGES A MEAL

To understand the rising scale, start with what happens at a meal.⁵ Semaglutide mimics GLP-1, a hormone released from the intestine that helps regulate blood sugar and appetite.^{5,3} During treatment, people can feel less hungry, more satisfied and better able to control eating.⁵ Researchers have tested those changes using something more concrete than a report of easier dieting: how much food people actually eat when given the chance.⁵

In one small experiment, 30 adults with obesity were assigned to weekly semaglutide and placebo in a randomly chosen order; 28 completed both periods.⁵ Each period lasted 12 weeks, and the semaglutide dose rose gradually to 1 mg.⁵ Because the same people experienced both

treatments, investigators could compare their eating under the two conditions.⁵

After a standard breakfast, participants were given lunch and dinner and told to eat until pleasantly full.⁵ An evening snack box extended the test beyond the main meals, offering sweet and savoury options that people could keep until midnight.⁵ Researchers measured consumption and found that the energy eaten across those freely chosen meals and snacks was about 24% lower on semaglutide (Figure 2).⁵ Nausea ratings were similar under both treatments, helping distinguish reduced appetite from simply feeling too sick to eat on the test day.⁵

The stomach is part of the story, but a permanently slow stomach is too simple an explanation.⁶ In another trial, 72 adults received semaglutide raised to the obesity dose or placebo for 20 weeks.⁶ At the end, a standard breakfast

included yoghurt containing paracetamol.⁶ Researchers measured absorption of that pain medicine as an indirect guide to how the meal left the stomach.⁶ Participants later ate as much lunch as they wanted.⁶ The semaglutide group ate about a third less, yet the indirect test found no evidence of delayed stomach emptying at that point.⁶ The test can miss delays, especially in solid food leaving the stomach.⁶ Even so, the pairing of lower food intake with this test result points to a fuller appetite story.⁶

What happens when that appetite effect is removed? How often appetite returns, and exactly when, remains uncertain.⁷ One small study followed a single group.⁷ It fol-

lowed 28 women with obesity through semaglutide treatment and 12 weeks afterward, reporting returning appetite in roughly 80%.⁷

The medicine itself leaves gradually.³ Its half-life is about a week: roughly half the remaining drug is eliminated over that interval.³ Stopping injections therefore allows drug levels to fall over weeks.³ A review of weight-loss biology describes another pressure toward regain: after losing weight, people can feel hungrier and sometimes burn less energy than expected even allowing for their smaller size.⁸ Those pressures form part of the background to stopping appetite treatment.⁸

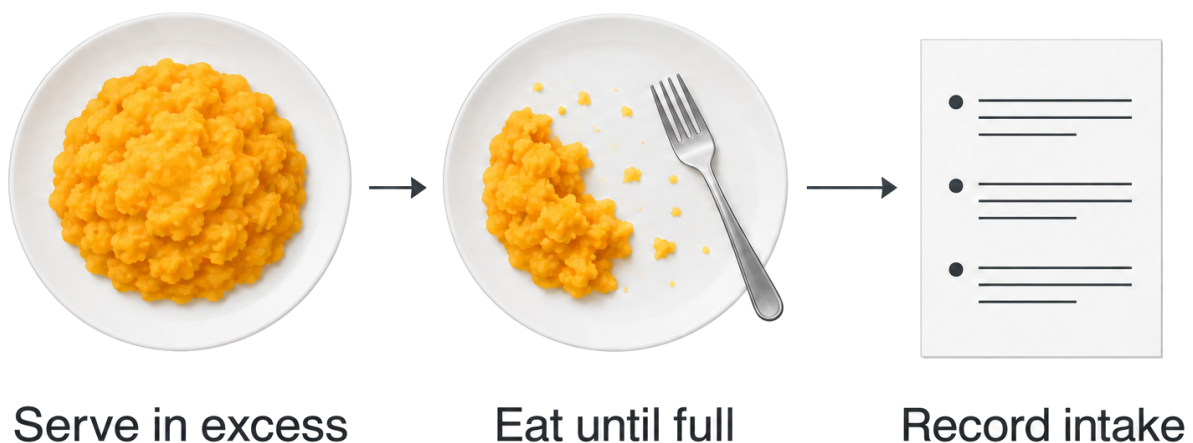


Figure 2. Appetite became a measurable meal. In the controlled tests, participants received food in excess, ate until comfortably full, and investigators recorded intake.^{5,6} The left-to-right sequence sketches that procedure. The pictured food is schematic; its composition and fraction left over carry no quantitative result.

03 THE SUPPORT STAYED, THE INJECTION CHANGED

The STEP 1 result leaves a puzzle: how much of the regain came from stopping medication, and how much from ending the counselling? STEP 4 changed the design so the support continued on both sides.² Adults without diabetes first spent 20 weeks taking semaglutide as the dose rose to 2.4 mg.² Investigators then assigned 803 people

who had reached that dose either to continue semaglutide or to switch to a matching dummy injection.²

Over the next 48 weeks, those continuing the medicine lost another roughly 8% of their weight, while those switched to placebo gained about 7%.² These changes were measured from the point of the switch, after the initial treatment loss (Figure 3).² The opposite directions emerged while both groups still received monthly professional counselling.² That comparison supplies

stronger evidence that removing semaglutide itself promotes regain, although it describes people who had already managed to stay on the target dose.²

The continuing support was specific: a daily food-energy target about 500 calories below estimated needs, 150 minutes of activity a week, and records of diet and activity reviewed at counselling visits.² Regain in the placebo group occurred within that programme, showing the limits of treating support as a guaranteed replacement for medication.²

There was a different trade-off in digestive symptoms.² New gastrointestinal events during the comparison period were reported by about four in ten people continuing semaglutide and about 26% of those switched to placebo.² This result describes new events over many months; it does not say when someone's existing nausea, vomiting or other digestive symptoms disappeared after stopping.² The STEP 1 follow-up did not systematically collect adverse events, leaving a gap precisely where readers might hope for a recovery calendar.¹ Here, “withdrawal” means stopping treatment; these studies did not test for an addiction-like withdrawal syndrome.^{1,2}

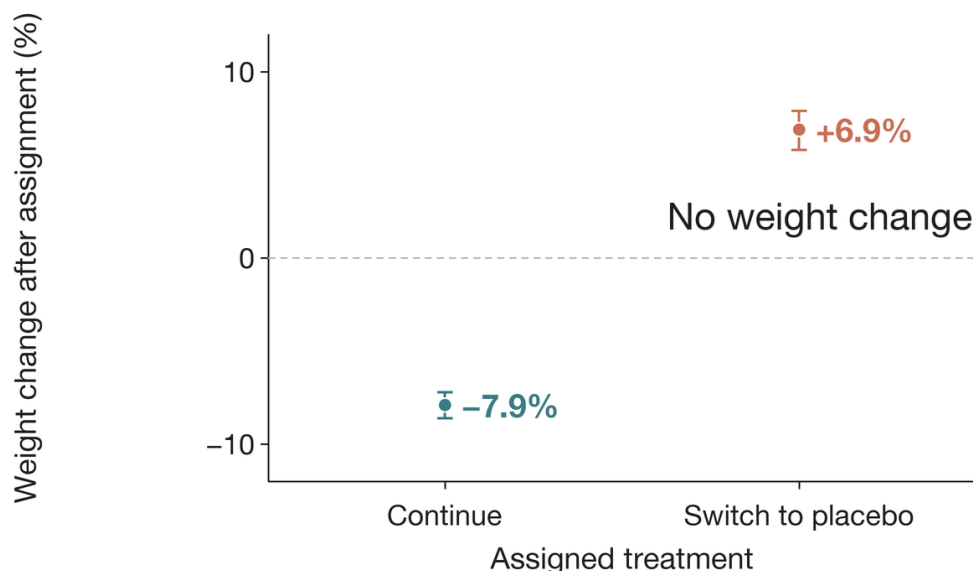


Figure 3. Continuing and stopping sent weight in opposite directions. The horizontal axis names assigned treatment; the vertical axis is percentage weight change from assignment, after the initial semaglutide treatment period. Dots show the group averages after 48 weeks and whiskers show 95% confidence intervals.^{2,9} The comparison included people who had reached 2.4 mg, with counselling continued in both groups.^{2,9}

04 BLOOD SUGAR HAS ITS OWN STORY

Weight is only one outcome of stopping a medicine developed for diabetes.³ Semaglutide helps the body release insulin and reduces release of glucagon, another hormone that regulates blood sugar.³ The stimulation of insulin release depends on glucose levels.³ Removing those effects can worsen glucose control, making weight trials in people without diabetes an incomplete guide to what stopping Ozempic means.¹⁰

A medicine shortage provided a window into that problem in South Korea.¹¹ Researchers examined records from adults with type 2 diabetes who had been taking dulaglutide, a related GLP-1 medicine, alongside metformin.¹¹ When dulaglutide became unavailable, some stopped without a replacement and many switched to other oral diabetes medicines.¹¹ The analysis included 69 people with usable blood tests before and about three months after the interruption.¹¹

Their average HbA1c rose by about a percentage point, and fasting blood glucose rose too.¹¹ HbA1c measures longer-term glucose exposure.¹² Researchers have com-

pared it with average glucose over the preceding three months, giving it a broader view than a single blood-sugar reading.¹² The shortage study was observational, used dulaglutide rather than Ozempic, and left out people who started insulin or lacked suitable tests, all of which narrow the result.¹¹

A broader review pooled four studies reporting HbA1c in diabetes and found an average rise of roughly 0.7 percentage points after GLP-1 treatment ended.¹⁰ It compared results at the end of treatment with those at least 12 weeks afterward.¹⁰ The studies used different medicines and follow-up care and their results varied greatly.¹⁰ Their combined figure supports worsening glucose control, with the size of the change for Ozempic users still uncertain.¹⁰

Other benefits can recede alongside glucose control.¹³ Further analyses of the same STEP 4 participants found that improvements in blood pressure and several blood-fat measures deteriorated after the switch to placebo.¹³ These measurements track risk factors.¹³ How they translate into actual heart attacks after stopping remains an open question.¹³ The body can move in several directions at once, with fewer new digestive events but less favourable weight and metabolic readings.^{2,13}

05 THE AVERAGE HAS EXCEPTIONS

Outside a trial, stopping can look less uniform.¹⁴ A study of nearly 8,000 adults in an Ohio and Florida health system followed people starting semaglutide or tirzepatide for obesity.¹⁴ Those who stopped lost less overall than those who continued, but their average weight trajectories were relatively stable after discontinuation during the available follow-up.¹⁴ That finding complicates the idea that every last injection starts the same steep climb.¹⁴

Many patients used lower doses and lost less weight than people in the major trials, which could help explain the contrast.¹⁴ “Stopping” meant a prescription-supply gap longer than about three months, sometimes followed by another refill.¹⁴ A year counted from starting treatment left late stoppers with much shorter follow-up afterward.¹⁴ Changes in diet or activity were not measured well enough to explain the difference.¹⁴

Partial retention also appeared in a much smaller study of women with obesity and polycystic ovary syndrome.¹⁵ They took semaglutide for 16 weeks while already receiving metformin, then continued metformin and lifestyle support for two years.¹⁵ The researchers reported regain of about a third of the semaglutide-associated loss, leaving

much of it retained.¹⁵ With only 25 women and no group stopping without metformin, the study cannot identify metformin as the reason.¹⁵

Together with the retained losses in STEP 1, these findings leave room for several outcomes after stopping.^{1,14,15} They have not established which combination of initial loss, continuing treatment and daily habits reliably predicts who will retain it.^{14,15}

06 KEEPING THE LOSS IS A SECOND EXPERIMENT

One maintenance experiment began before anyone's final injection.¹⁶ In Denmark, adults with obesity first completed an eight-week low-calorie programme using four meal-replacement products a day.¹⁶ Researchers then assigned 195 participants to a year of supervised exercise, daily liraglutide—a different GLP-1 medicine—the combination, or placebo.¹⁶ After that year, the active treatments ended and participants were invited back a year later.¹⁶

Exercise had meant vigorous indoor cycling followed by circuit training, with participants encouraged to attend group sessions twice weekly and exercise individually twice weekly.¹⁶ When supervision ended, they returned the sports watches and heart-rate monitors used in the programme.¹⁶ The follow-up asked what remained after that organised support, or the medicine, was removed.¹⁶

People previously assigned liraglutide alone regained about 10 kg during the off-treatment year, compared with about 4 kg after exercise alone.¹⁶ Regain after the combined programme was about 7 kg, but its difference from liraglutide alone was uncertain.¹⁶ Only 109 participants returned, and some differences failed a stricter check for chance findings across multiple comparisons.¹⁶ Exercise remains a plausible part of maintenance, with those limits and the different medicine leaving a semaglutide exit strategy unproven.¹⁶

Current guidance reflects the stronger evidence for continuing effective treatment.⁹ The 2025 Canadian adult obesity guideline recommends long-term use of effective obesity medication alongside changes in daily habits to sustain weight loss and avoid regain and the loss of health benefits.⁹ The programmes behind that discussion involve repeated food and activity counselling, records reviewed with professionals, or supervised exercise.^{2,16}

Tapering remains a question awaiting a completed comparison.¹⁷ A published semaglutide trial protocol proposes

testing a gradual reduction over 16 weeks against immediate stopping, but it has yet to report results.¹⁷ Reviews describe reducing doses or spacing injections as possible maintenance approaches with limited supporting evidence.⁸ A reliable general schedule has not emerged from these sources.^{8,17}

The appetite and blood-sugar effects of semaglutide depend on treatment that eventually leaves the body.³ Stopping commonly brings substantial weight regain, while some people retain part of the loss.¹ The open question is which forms of continuing care can keep more of those benefits when the injections end; the proposed tapering comparison explores one possible route.^{8,17}

07 THE NUMBERS BEHIND THE STORY

STUDY	PEOPLE AND COMPARISON	RESULT	FOLLOW-UP
STEP 1 extension	327, both original arms; all stopped	Semaglutide: 17.3% loss, 11.6 percentage-point regain, 5.6% net loss ¹	52 weeks off
STEP 1 retained loss	197 former semaglutide participants	95/197 (48.2%) remained ≥5% below baseline ¹	52 weeks off
Crossover meal study	30 assigned, 28 completed; semaglutide/placebo	24% less freely eaten energy on semaglutide ⁵	12 weeks/treatment
STEP 4 weight	803; continuation/placebo, counselling both	−7.9%/+6.9% from assignment; difference −14.8 percentage points (95% confidence interval −16.0 to −13.5) ²	48 weeks
STEP 4 digestive events	535 continuation, 268 placebo	New gastrointestinal events: 41.9%/26.1% ²	48 weeks
Diabetes withdrawal review	Four HbA1c studies; 219 participants	HbA1c +0.65 percentage points (95% confidence interval 0.22–1.08) ¹⁰	≥12 weeks off
Korean shortage cohort	69; dulaglutide cessation	HbA1c 7.0% before, 8.1% afterward ¹¹	Three months
Danish exercise follow-up	195 assigned; 109 returned	Regain: liraglutide 9.6 kg, exercise 3.6 kg, combination 7.1 kg ¹⁶	One year off

The STEP 1 percentages are separately reported study results; rounding means they need not subtract exactly.

Scope and methods

This narrative review covers human research on stopping semaglutide, with related GLP-1 medicines explicitly identified. Literature searches on 5 October 2026 used PubMed and citation searching, including studies that

complicate the main regain finding. Crossref was used to verify references. Animal experiments and claims about a personal symptom schedule or the proportions of muscle and fat regained were excluded. The small appetite-return study was available only as an abstract. The tapering protocol supplies no completed outcome results.¹⁷

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